English speakers learn to use F1, disregard F2 in Arabic perception

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CLS 62

Background

Cross-linguistically, F2 transitions in CV sequences are cues to consonant place of articulation and are used in listener perception.

Native Arabic speakers use F1, not F2, as a primary cue for uvular and pharyngeal consonant contrasts (Alwan, 1989).

F2 is the primary place cue in English (Delattre et al., 1955).

This study extends Alwan (1989) to L1 English L2 Arabic learners: **Can English speakers use F1 information and ignore F2?**

Hypotheses

Current models (SLM-r, L2LP), predict that L2 learners can develop native-like cue weighting (Flege & Bohn, 2021; Escudero, 2005).

1. **Cue weighting can be learned in the L2; learners will rely on L2 cues more than less informative L1 cues.**

Learning new cues is very difficult, even after long L2 exposure: L1 Japanese speakers are not sensitive to F3 in perception of L2 English /r/-/l/ contrast (Miyawaki et al., 1975; Takagi Mann, 1995).

Previous work has found only general advantages of L2 experience in L3 perception (Onishi, 2018).

Mashaqba et al. (2022) suggest there are production advantages for L3 Arabic learners who have L2 experience with guttural consonants.

2. **Learners with relevant L2 experience will be more native-like than learners with no relevant experience.**

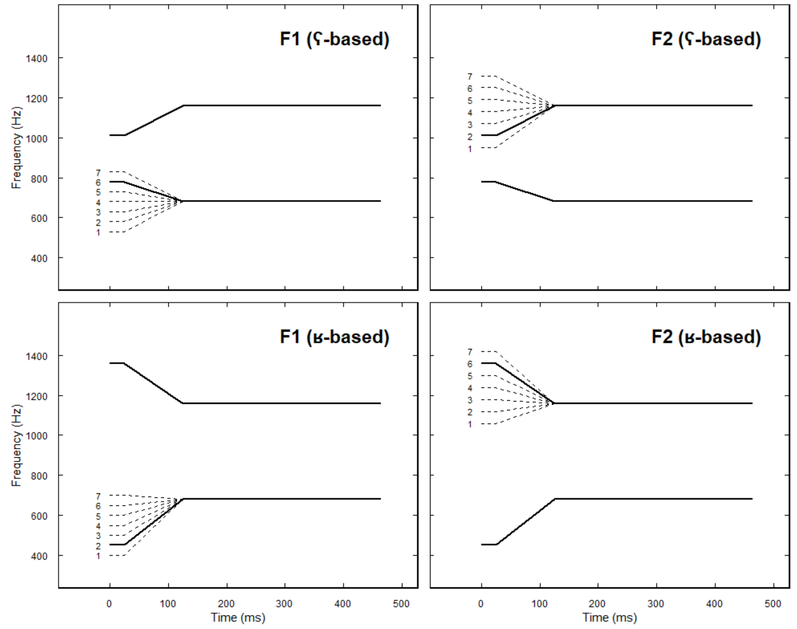
Other	Post-velar
Greek (1)	Farsi (1): post-velar ɣ, ʔ, h
Lithuanian (1)	French (8): ʁ
Mandarin (4)	Hindi-Urdu (3): post-velar ɣ, ɦ, q, h
Spanish (14)	Japanese (1): ɴ
Swedish (1)	
No prior L2 (3)	

[1] Relevant phonemes in L2 languages spoken by participants.

Learners who have experience with L2 guttural consonants will have perceptual advantages in L3 Arabic over learners without experience: Post-velar experience learners will use more F1 than other learners.

Stimuli

Seven-step continua with manipulated onset values of either F1 or F2. Replicated from Alwan (1989) using the Klatt synthesizer in Praat.



[2] Manipulated vowel onset transitions for ʕ- (top) and ʁ-based continua (bottom).

Methodology

Participants: 5 L1 Arabic, 15 beginner L2, 15 advanced L2.
(<1 sem) (>1 year)

Task: Three-alternative forced choice between /ʕ, ʁ, ʔ/.
28 stimuli x 6 repetitions.

Measure: Mutual information (MI) is the amount of information shared between two variables. Higher MI corresponds to increasing dependence.

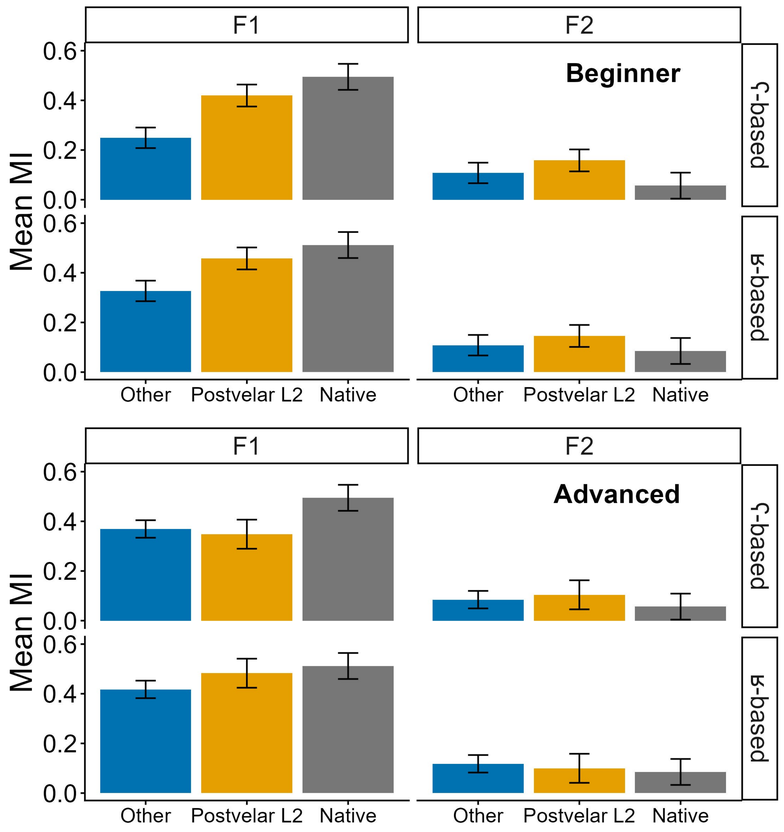
$$MI(response; stimulus) = H(response) - H(response|stimulus)$$

Selected References

Alwan, A. (1989). Perceptual cues for place of articulation for the voiced pharyngeal and uvular consonants. Journal of the Acoustical Society of America, 86(2), 549–556. **Delattre, P. C., Liberman, A. M., Cooper, F. S. (1955).** Acoustic loci and transitional cues for consonants. The Journal of the Acoustical Society of America, 27 (4), 769–773. **Mashaqba, B., Huneety, A., Guba, M. A., & Al-Duneibat, B. (2022).** Production of gutturals by non-native speakers of Arabic. Indonesian Journal of Applied Linguistics, 12(2), 334–347. **Onishi, H. (2018).** The effects of L2 experience on L3 perception. In Advances in the investigation of L3 phonological acquisition (pp. 65–81). Routledge.

Contact and further information at: https://aherlant.github.io/

Results



[3] Use of F1 and F2 by length and type of learner post-velar experience (+/- 1 SE)

Findings

1. Arabic learners use more F1 information than F2 ($p < .0001$).
Learning happens quickly, within 5 months of exposure.
Proposal: repurposing an existing cue for a new dimension is easier than learning to use an entirely new cue (eg. use of F3 by L1 Japanese L2 English speakers).

2. Beginner learners without post-velar experience use less F1 than learners with experience ($p < .05$).
Exposure to specific L2 sound inventories can lead to different learning trajectories in the L3, but effect limited to beginners.
L2 guttural knowledge generalizes to new sounds in L3 Arabic.